

MACAD 2025/26 · DIGITAL TOOLS FOR GRAPH MACHINE LEARNING

GraphML_Go2

Graph Learning Pipeline for Architectural Floor Plans

FINAL ASSIGNMENT

IAAC · MaCAD 2025/26

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Contents

Dataset exploration, research context, graph structure, building selection, and first results from graph analysis on the Narkomfin Building.

01 Dataset

Modified Swiss Dwellings: 5,372 residential floor plans as graphs

02 Research

MSD benchmark paper (ECCV 2024) · GNN node classification paper (eCAADe 2024)

03 Graph Structure

graph_in vs graph_out: zone labels, connectivity, and room type prediction

04 Pipeline

Floor plan design → graph construction & analysis → node classification

05 The Narkomfin Building

Building introduction · spatial concept · why it maps to graph ML

01 DATASET

Modified Swiss Dwellings

- 5,372 floor plans
- Multi-apartment clusters
- Corridor buildings
- Single units
- Graph modality only



5,372
Floor Plans

4
Zone Types

9
Room Types

.pickle
Graph Format

02 RESEARCH

Literature

PAPER 1

MSD: A Benchmark Dataset for Floor Plan Generation of Building Complexes

ECCV 2024 · TU Delft

- 5,372 Swiss residential floor plans
- 3 modalities: image, geometry, graph
- Richer than RPLAN & LIFULL: multi-apartment, compass orientation, inter-unit links
- Graph encodes spatial zones, not room types

This dataset is our primary data source.

PAPER 2

GNN for Node Classification and Attribute Allocation in Architectural BIM

eCAADe 2024 · Cardiff University

- Uses the same MSD dataset
- Converts floor plans to graphs via TopologicPy
- Benchmarks GCN, GAT, GraphSAGE architectures
- GraphSAGE-Pool, 4 layers, ~95% accuracy

The pretrained model from this paper is what we will run on our floor plan.

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GRAPH STRUCTURE

Data Structure

graph_in: model input

Attribute	Type	Values
zoning_type	int	0 · 1 · 2 · 3
connectivity	str	'door' · 'entrance'

graph_out: ground truth

Attribute	Type	Description
room_type	int	0 – 8
geometry	Polygon	2D room outline
centroid	Point	centre of room

Room types (0 – 8)

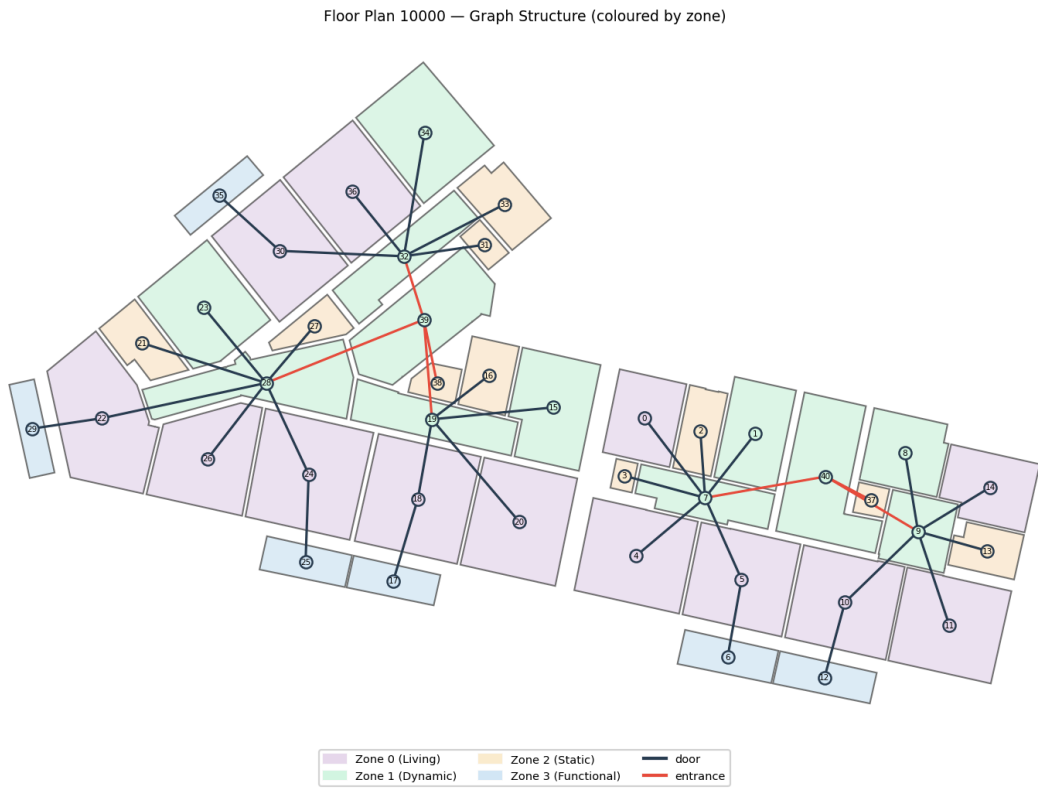
ID	Room	ID	Room
0	Balcony	5	Kitchen
1	Bathroom	6	Living Room
2	Bedroom	7	Storeroom
3	Corridor	8	Other
4	Dining		

Zone types (0 – 3)

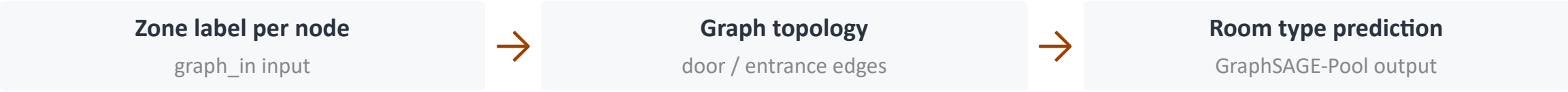
ID	Zone	Typical rooms
0	Living	Living room, dining
1	Dynamic	Corridor, entrance
2	Static	Bedroom, bathroom
3	Functional	Kitchen, storeroom

03 GRAPH STRUCTURE

From Plan to Graph: Plan 10000



Plan 10000: room types (left) and spatial connectivity graph (right)



04 NEXT STEPS

Pipeline

Floor Plan Input

- 01** Use the F-type and K-type floor plans from the Narkomfin Building as our input geometry, redrawn in Rhino as closed polylines with shared edges for TopologicPy adjacency detection.

Graph Construction & Analysis / TopologicPy

- 02** Convert geometry to a graph, label nodes with `zoning_type` and edges as door/entrance, then compute degree, betweenness, and closeness centrality.

Node Classification / GraphSAGE-Pool

- 03** Run the pretrained model to predict `room_type` from zone labels and connectivity. Compare results against MSD ground truth.

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OUR BUILDING

The Narkomfin Building



Post-restoration render · Ginsburg Architects · Moscow, 2021

ARCHITECTS	YEAR / LOCATION	TYOLOGY	SCALE	UNIT TYPES
Moisei Ginzburg & Ignaty Milinis	1930 · Moscow, Russia	Collective housing, dom- kommuna	54 apartments · 6 floors	F-type duplex · K- type communal

A landmark of Soviet Constructivism, designed as a **social condenser**: private space minimised, shared infrastructure maximised.

Two distinct floor plan typologies and a clear spatial hierarchy make it an ideal subject for graph-based analysis.

05 OUR BUILDING

Why Narkomfin

SPATIAL CONCEPT

The Social Condenser

- Private cells minimal: sleep and work only
- Shared corridor spine: the sole circulation artery
- Communal programme in a separate linked block
- Legible spatial hierarchy, ideal for graph analysis

WHY IT FITS GRAPH ML

Clear Zoning · Two Plan Types

- Corridor → **Dynamic (1)**, unambiguous node label
- Cells → **Static (2)**, communal → **Living (0) + Functional (3)**
- F-type & K-type units: two spatial configurations to compare
- Orthogonal geometry → clean TopologicPy graph construction

Soviet Constructivism · 1930

Social condenser

Linear spine

F-type duplex

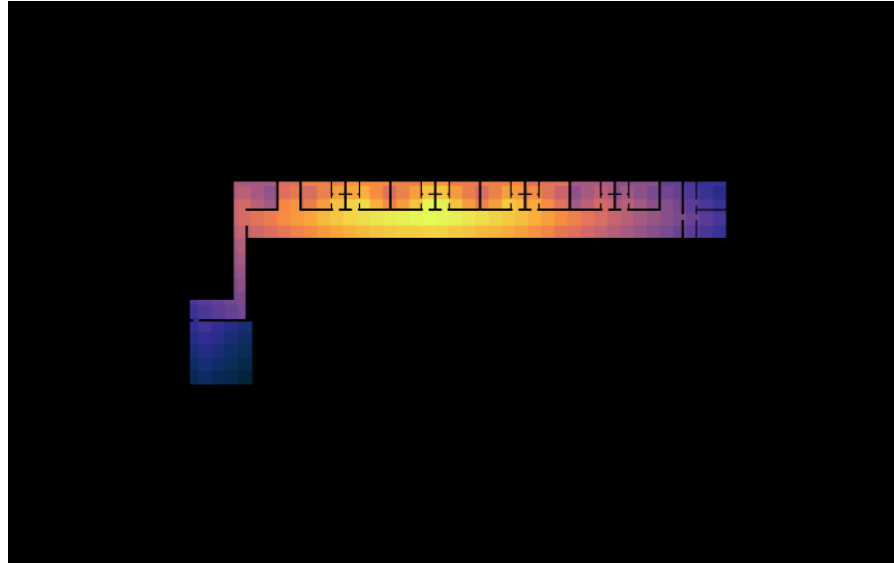
K-type communal

Modular repetition

06

GRAPH ANALYSIS

Plan 1 — Closeness Centrality



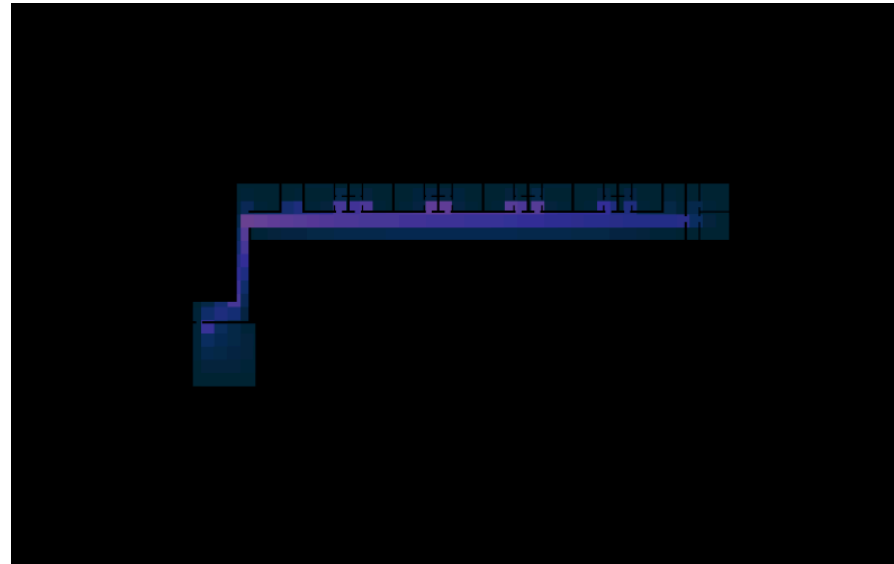
Thermal scale — warm = high integration, cool = low integration

The corridor is the most spatially integrated element in the plan. Its warm orange and yellow tones confirm it as the global connector from which every room is reachable in the fewest steps. Closeness drops progressively as you move into the private cells. The staircase block at the far end scores the lowest, functioning as a terminal dead-end rather than a through-space. This gradient directly reflects the building's design intent: the corridor is not incidental infrastructure but the social and circulatory core of the plan.

06

GRAPH ANALYSIS

Plan 1 — Betweenness Centrality



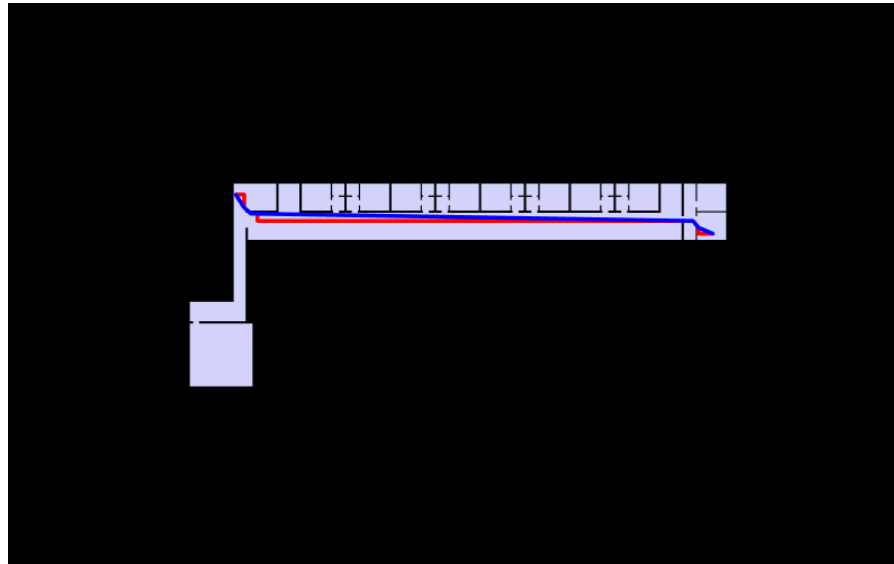
Thermal scale — warm = high traffic, cool = low traffic

Betweenness centrality reveals which spaces lie on the most shortest paths across the plan. The central corridor sections score highest — any journey from one end of the building to the other must pass through them. Rooms at the ends have near-zero betweenness: they are destinations, not conduits. The staircase block again registers low, consistent with its peripheral role. The result confirms the corridor as both the most accessible and the most traversed spine — a spatial double function that the Constructivist brief made explicit.

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GRAPH ANALYSIS

Plan 1 — Shortest Path



Grid path (red, 77.2 units) vs straightened path (blue, 74.4 units)

The shortest path from the upper-left corner to the lower-right runs almost entirely along the corridor. The grid-based path (red, 77.2 units) and the geometrically straightened path (blue, 74.4 units) are nearly identical — straightening recovers only 3.7% in distance. This near-coincidence is a direct consequence of the linear plan: the corridor already is the straight line, so there is little room for a shortcut. The graph confirms what the architecture embodies — in the Narkomfin Building, the most efficient route and the only social route are the same.